MINAHIL MOBIN

MS Cybersecurity Student • AI/ML • Generative AI • Cloud Security
Lahore, Pakistan • GitHub • LinkedIn • Portfolio

Education

MS Cybersecurity (In Progress) 2025 - 2027
National University of Computer and Emerging Sciences (FAST-NUCES), Lahore CGPA: 3.17

BS Data Science 2021 - 2025
National University of Computer and Emerging Sciences (FAST-NUCES), Lahore

Experience

AI/ML Intern, Abacus July 2026 - August 2026

- Developed Python applications using object-oriented programming, file handling, exception handling, and modular design.
- Built automation tools for document processing, text analysis, and JSON report generation.
- Performed data cleaning, exploratory data analysis (EDA), and visualization using Pandas and Matplotlib.
- Implemented and evaluated machine learning models including Linear Regression, Logistic Regression, Decision Trees, and K-Means using Scikit-learn.

MIS Intern, Shaukat Khanum Memorial Cancer Hospital June 2024 - July 2024

- Researched healthcare information systems and emerging AI applications in clinical environments.
- Explored the use of Large Language Models (LLMs) for clinical documentation assistance and medical information retrieval.
- Evaluated healthcare datasets and reviewed current research to assess their suitability for machine learning applications.

Selected Academic Projects

Explainable AI-Based Intrusion Detection System for Cloud Security May 2026

- Developed an Explainable AI-based Intrusion Detection System (IDS) using Decision Tree, Random Forest, and SVM classifiers on the CICIDS2017 dataset.
- Selected Random Forest as the best-performing model and deployed it as a Flask REST API on AWS EC2 for real-time intrusion detection.
- Integrated SHAP for model interpretability and evaluated robustness through cross-dataset and noise-based testing, highlighting generalization challenges in cloud environments.

Feature Selection-Based Lightweight Intrusion Detection System May 2026

- Designed a lightweight Random Forest-based IDS using Mutual Information feature selection on the CICIDS2017 dataset.
- Evaluated seven feature subsets and achieved 99.82% classification accuracy using only 15 of 78 network traffic features (80.8% feature reduction).
- Analyzed the trade-off between feature reduction, model complexity, and detection performance for efficient intrusion detection.

Dyslexia Support Hub (AI Learning Platform) August 2024 - May 2025

- Developed a full-stack AI-assisted learning platform using React and Django to support personalized education for children with dyslexia.
- Built adaptive learning modules and integrated machine learning techniques to generate personalized learning recommendations based on student progress.

Inventory Management System February 2023 - May 2023

- Developed a full-stack inventory management system using React and JavaScript for efficient product tracking and management.
- Implemented CRUD operations, input validation, and structured workflows to improve data management and system reliability.
- Managed project planning, documentation, and development workflow throughout the software lifecycle.

Technical Skills

Programming: Python, JavaScript, SQL, C++
AI/ML: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, SHAP, Model Evaluation
Generative AI: LLMs, RAG, Prompt Engineering, Embeddings
Cybersecurity: IDS, Network Security, Cloud Security, Security Analytics, Wireshark
Tools: React, Django, REST APIs, AWS EC2, MySQL, PostgreSQL, Git

Certifications

AWS Academy Graduate - Cloud Security Builder May 2026
AWS Academy Graduate - Cloud Security Foundations April 2026
AWS Academy Graduate - Cloud Foundations April 2026